

SENG1050

Lab 3

In this lab, you will gain experience in creating valid XML documents with DTD.

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Task 1 - Creating XML Data

1. Look at the following valid XML document `bibliography.xml`.

```
<?xml version="1.0" encoding="utf-8"?>
<!DOCTYPE bibliography [
<!ELEMENT bibliography (book+)>
  <!ELEMENT book (author+, title, pub?, year, URL*)>
    <!ELEMENT author (
      (givenname | nickname)?, surname)>
      <!ELEMENT givenname (#PCDATA)>
      <!ELEMENT nickname (#PCDATA)>
      <!ELEMENT surname (#PCDATA)>
      <!ELEMENT title (#PCDATA)>
      <!ELEMENT pub (#PCDATA)>
      <!ELEMENT year (#PCDATA)>
      <!ELEMENT URL (#PCDATA)>
    ]>
<bibliography>
  <book>
    <author>
      <nickname>Fred</nickname>
      <surname>Flintstone</surname>
    </author>
    <author>
      <givenname>Helen</givenname>
      <surname>James</surname>
    </author>
    <title>"Statistical Database Disclosure"</title>
    <year>2006</year>
    <URL>http://some.webaddress.com</URL>
    <URL>http://another.webaddress.com</URL>
  </book>
  <book>
    <author>
      <surname>Simpson</surname>
    </author>
    <title>"Homer's Odyssey"</title>
    <pub>Springer Verlag</pub>
    <year>2003</year>
  </book>
</bibliography>
```

2. Make sure you understand how the included DTD works in defining the structure of the XML. You may find it useful to draw an XML tree, as shown in the lecture notes, to help you visualise the structure.
3. In addition to a book entry (which has already been done), try to modify the DTD section to include information for a journal article. The bibliography entry for a journal article contains the following information: author(s) names (given name, surname, nickname, article title, journal name, volume number, issue number, page numbers, year, URL).
4. Then, modify the XML section to include at least one journal article after the books.
5. Validate your XML against the DTD by submitting it to an [XML Validator](#).
Note: Sometimes this website is overloaded and doesn't work correctly. If this occurs, you can use [this XML Validator](#) instead. However, a free account sign up is required to check external DTD. Also, it is NOT recommended to use other validators, such as the one from W3 Schools, which only check well-formedness and exclude DTD checking.

Task 2 - Creating XML Data

1. Create a new plain-text file called `airline.xml`.
2. Using your favourite text editor, copy the following information (representing one "booking") into `airline.xml`.

```
Passenger: Paul J. Franks
Address: University of Newcastle, University Drive, Callaghan, 2308
Phone: 0249216175
Leg:
  Date: 11 Aug 2003
  Flight: QF005
  Depart: SYD, 15:45
  Arrive: SIN, 21:35
  Class: Economy
  Aircraft: B744
Leg:
  Date: 21 Aug 2003
  Flight: QF010
  Depart: SIN, 19:45
  Arrive: MEL, 04:55
  Class: Economy
Leg:
  Date: 24 Aug 2003
  Flight: QF434
  Depart: MEL, 13:00
  Arrive: SYD, 14:20
  Class: Business
  Aircraft: B73H
```

3. Add tags to make `airline.xml` a *well-formed XML document*.
 - The root tag should be `<bookings>`.
 - All the booking information for one passenger should be enclosed in a `<booking>` tag.
 - While adding tags, give some consideration to the structure of the data. For example, a passenger name consists of three parts: family name, given name, and initials, whereas departure and arrival data consists of an airport code and a time.
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Task 3 - Create a Document Type Definition

1. Add an internal Document Type Definitions (DTD) to `airline.xml`. Ensure these rules are imposed:
 - There are potentially many (but possible zero) bookings.
 - Addresses and phone numbers are optional.
 - There is at least one leg in each bookings.
 - Class should be made an optional attribute of a leg
 - Aircraft information is optional.
 - Create two entities, one to convert B744 to 'Boeing 744', and one to convert B73H to 'Boeing 737-800'. Use these entities in the XML.
 2. Validate your XML against the DTD by submitting it to an [XML Validator](#).
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Task 4 - Create DTD and XML File for Data Transfer

(To be completed as homework if there is not sufficient time in the laboratory.)

Coles has hired you to transfer all of its customer records from the Oracle database to MySQL database. You decide to use XML for transferring the records. The following information is stored for each customer: name, date of birth, address, phone number, Flybuys number, and a sequence of purchase records. In each purchase record, the stored information is: purchase date, purchase time, store ID, total amount, and the list of items purchased. For each purchased item, the following information is saved: item code, item name, unit price, quantity, and item price. The following rules are imposed:

- Each customer record has potentially many (but possible zero) purchase records.

- Address of customer is optional but phone number is mandatory.
- There should be at least one item in each purchase.

Your first step is to get the following steps done for data transfer:

1. Create an external DTD for representing the customer purchase history.
 2. Create a valid XML file with at least two customer records where each customer record contains at least two purchase records.
 3. Validate your XML against the DTD by submitting it to an [XML Validator](#).
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